```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sea
```

```
#loading the dataset
```

```
data = pd.read_csv("imdb movies data.csv")
```

```
data.head()
```

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
0		NaN	NaN	Drama	NaN	NaN	J.S. Randhawa	Manmauji	Birbal	Rajendra Bhatia
1	#Gadhvi (He thought he was Gandhi)	-2019.0	109 min	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
2	#Homecoming	-2021.0	90 min	Drama, Musical	NaN	NaN	Soumyajit Majumdar	Sayani Gupta	Plabita Borthakur	Roy Angana

Next steps: Generate code with data New interactive sheet

```
data.shape
```

```
(15509, 10)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15509 entries, 0 to 15508
Data columns (total 10 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Name      15509 non-null  object
 1   Year      14981 non-null  float64
 2   Duration  7240 non-null   object
 3   Genre     13632 non-null  object
 4   Rating    7919 non-null   float64
 5   Votes     7920 non-null   object
 6   Director  14984 non-null  object
 7   Actor 1   13892 non-null  object
 8   Actor 2   13125 non-null  object
 9   Actor 3   12365 non-null  object
dtypes: float64(2), object(8)
memory usage: 1.2+ MB
```

```
data.describe(include='all')
```

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
count	15509	14981.000000	7240	13632	7919.000000	7920	14984	13892	13125	12365
unique	13838	NaN	182	485	NaN	2034	5938	4718	4891	4820
top	Mamta	NaN	120 min	Drama	NaN	8	Jayant Desai	Ashok Kumar	Rekha	Pran
freq	7	NaN	240	2780	NaN	227	58	158	83	91
mean	NaN	-1987.012215	NaN	NaN	5.841621	NaN	NaN	NaN	NaN	NaN
std	NaN	25.416689	NaN	NaN	1.381777	NaN	NaN	NaN	NaN	NaN
min	NaN	-2022.000000	NaN	NaN	1.100000	NaN	NaN	NaN	NaN	NaN
25%	NaN	-2009.000000	NaN	NaN	4.900000	NaN	NaN	NaN	NaN	NaN
50%	NaN	-1991.000000	NaN	NaN	6.000000	NaN	NaN	NaN	NaN	NaN
75%	NaN	-1968.000000	NaN	NaN	6.800000	NaN	NaN	NaN	NaN	NaN
max	NaN	-1913.000000	NaN	NaN	10.000000	NaN	NaN	NaN	NaN	NaN

```
data.isnull().sum()
```

	0
Name	0
Year	528
Duration	8269
Genre	1877
Rating	7590
Votes	7589
Director	525
Actor 1	1617
Actor 2	2384
Actor 3	3144

dtype: int64

```
#maximum columns are having null values
```

```
#so we need to remove all that columns which are having too much null values and
#The column which is not critical for our prediction
```

```
data.drop(columns=['Duration'], inplace=True)
```

```
# Droping rows where 'Rating' is missing
data = data[data['Rating'].notnull()].copy()
```

```
data.reset_index(drop=True, inplace=True)
```

```
data.drop(columns='Name', inplace=True)
```

```
# Convert Year to positive values by multiplying by -1
data['Year'] = data['Year'].apply(lambda x: -x if x < 0 else x)
```

```
# Removeing commas and convert to float
data['Votes'] = data['Votes'].str.replace(',', '', regex=False).astype(float)

# Fill missing votes with median
data['Votes'].fillna(data['Votes'].median(), inplace=True)
```

```
/tmp/ipykernel_1519/3302237563.py:5: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chain
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are sett

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df


  data['Votes'].fillna(data['Votes'].median(), inplace=True)
```

```
#We'll fill nulls in categorical columns (Genre, Director, Actor 1, Actor 2, Actor 3) with "Unknown":

cols_to_fill = ['Genre', 'Director', 'Actor 1', 'Actor 2', 'Actor 3']

for col in cols_to_fill:
    data[col].fillna('Unknown', inplace=True)
```

```
/tmp/ipykernel_1519/2707083502.py:6: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chain
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are sett

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df


  data[col].fillna('Unknown', inplace=True)
```

```
data.isnull().sum()
```

	0
Year	0
Genre	0
Rating	0
Votes	0
Director	0
Actor 1	0
Actor 2	0
Actor 3	0

dtype: int64

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7919 entries, 0 to 7918
Data columns (total 8 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Year      7919 non-null   float64
 1   Genre     7919 non-null   object
 2   Rating    7919 non-null   float64
 3   Votes     7919 non-null   float64
 4   Director  7919 non-null   object
 5   Actor 1   7919 non-null   object
 6   Actor 2   7919 non-null   object
 7   Actor 3   7919 non-null   object
dtypes: float64(3), object(5)
memory usage: 495.1+ KB
```

```
data['Genre']
```

	Genre
0	Drama
1	Comedy, Romance
2	Comedy, Drama, Musical
3	Drama, Romance, War
4	Horror, Mystery, Thriller
...	...
7914	Action, Crime, Drama
7915	Action, Crime, Drama
7916	Action
7917	Action, Drama
7918	Action, Drama

7919 rows × 1 columns

dtype: object

```
#Split the Genre strings into lists
```

```
data['Genre'] = data['Genre'].str.split(', ')
data['Genre']
```

	Genre
0	[Drama]
1	[Comedy, Romance]
2	[Comedy, Drama, Musical]
3	[Drama, Romance, War]
4	[Horror, Mystery, Thriller]
...	...
7914	[Action, Crime, Drama]
7915	[Action, Crime, Drama]
7916	[Action]
7917	[Action, Drama]
7918	[Action, Drama]

7919 rows × 1 columns

dtype: object

```
# We explode the list so each genre gets its own row temporarily, then create binary columns using get_dummies()
```

```
genre_dummies = data['Genre'].explode().str.get_dummies().groupby(level=0).sum()
```

```
print(genre_dummies)
```

```
      Action  Adventure  Animation  Biography  Comedy  Crime  Documentary  \
0          0          0          0          0       0      0            0
1          0          0          0          0       1      0            0
2          0          0          0          0       1      0            0
3          0          0          0          0       0      0            0
4          0          0          0          0       0      0            0
...      ...        ...        ...        ...     ...    ...          ...
7914       1          0          0          0       0      1            0
7915       1          0          0          0       0      1            0
7916       1          0          0          0       0      0            0
7917       1          0          0          0       0      0            0
7918       1          0          0          0       0      0            0

      Drama  Family  Fantasy  ...  Musical  Mystery  News  Romance  Sci-Fi  \
0         1       0        0  ...        0        0     0        0       0
1         0       0        0  ...        0        0     0        1       0
2         1       0        0  ...        1        0     0        0       0
3         1       0        0  ...        0        0     0        1       0
4         0       0        0  ...        0        1     0        0       0
...     ...     ...      ...  ...      ...      ...   ...      ...     ...
7914      1       0        0  ...        0        0     0        0       0
7915      1       0        0  ...        0        0     0        0       0
7916      0       0        0  ...        0        0     0        0       0
7917      1       0        0  ...        0        0     0        0       0
7918      1       0        0  ...        0        0     0        0       0

      Sport  Thriller  Unknown  War  Western
0         0         0        0    0        0
1         0         0        0    0        0
2         0         0        0    0        0
3         0         0        0    1        0
4         0         1        0    0        0
...     ...       ...      ...  ...      ...
7914      0         0        0    0        0
7915      0         0        0    0        0
7916      0         0        0    0        0
7917      0         0        0    0        0
7918      0         0        0    0        0

[7919 rows x 23 columns]
```

```
data= pd.concat([data, genre_dummies], axis=1)
```

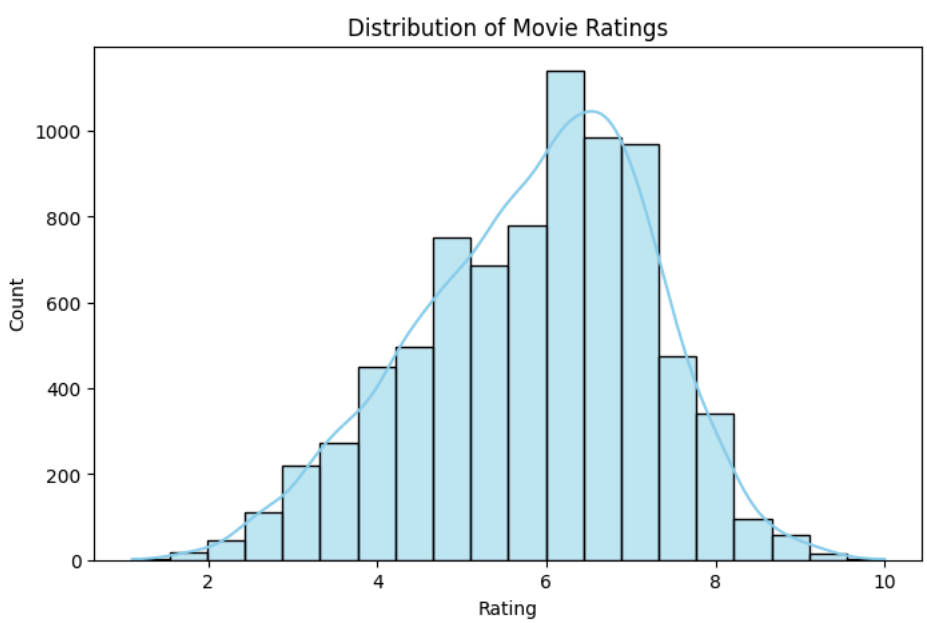
```
data.drop(columns='Genre', inplace=True)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7919 entries, 0 to 7918
Data columns (total 30 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Year         7919 non-null   float64
 1   Rating       7919 non-null   float64
 2   Votes        7919 non-null   float64
 3   Director     7919 non-null   object
 4   Actor 1      7919 non-null   object
 5   Actor 2      7919 non-null   object
 6   Actor 3      7919 non-null   object
 7   Action       7919 non-null   int64
 8   Adventure    7919 non-null   int64
 9   Animation    7919 non-null   int64
 10  Biography    7919 non-null   int64
 11  Comedy       7919 non-null   int64
 12  Crime        7919 non-null   int64
 13  Documentary  7919 non-null   int64
 14  Drama        7919 non-null   int64
 15  Family       7919 non-null   int64
 16  Fantasy      7919 non-null   int64
 17  History      7919 non-null   int64
 18  Horror       7919 non-null   int64
 19  Music        7919 non-null   int64
 20  Musical      7919 non-null   int64
 21  Mystery      7919 non-null   int64
 22  News         7919 non-null   int64
 23  Romance      7919 non-null   int64
 24  Sci-Fi       7919 non-null   int64
 25  Sport        7919 non-null   int64
 26  Thriller     7919 non-null   int64
 27  Unknown      7919 non-null   int64
 28  War          7919 non-null   int64
 29  Western      7919 non-null   int64
dtypes: float64(3), int64(23), object(4)
memory usage: 1.8+ MB
```

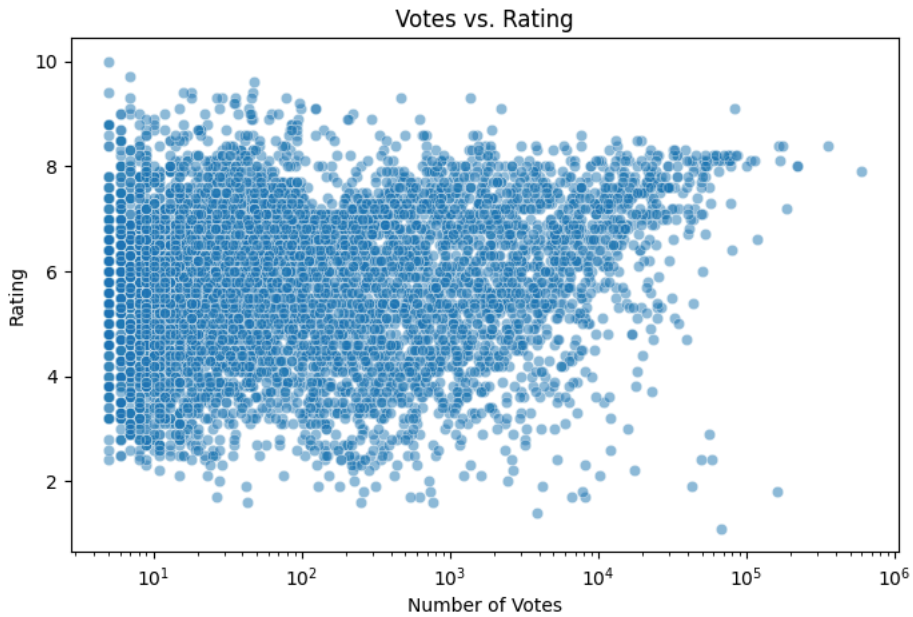
```
# DATA ANALYSIS EDA
```

```
plt.figure(figsize=(8, 5))
sea.histplot(data['Rating'], bins=20, kde=True, color='skyblue')
plt.title('Distribution of Movie Ratings')
plt.xlabel('Rating')
plt.ylabel('Count')
plt.show()
```

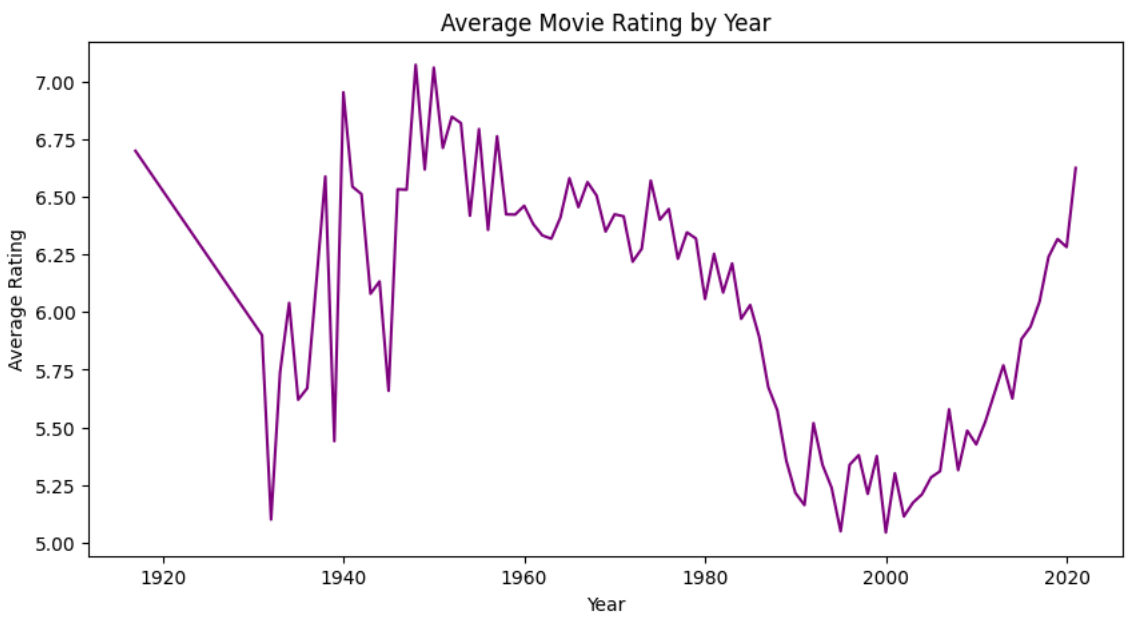



```
plt.figure(figsize=(8, 5))
sea.scatterplot(x='Votes', y='Rating', data=data, alpha=0.5)
plt.title('Votes vs. Rating')
plt.xlabel('Number of Votes')
```

```
plt.ylabel('Rating')
plt.xscale('log')  # Use log scale to handle outliers
plt.show()
```



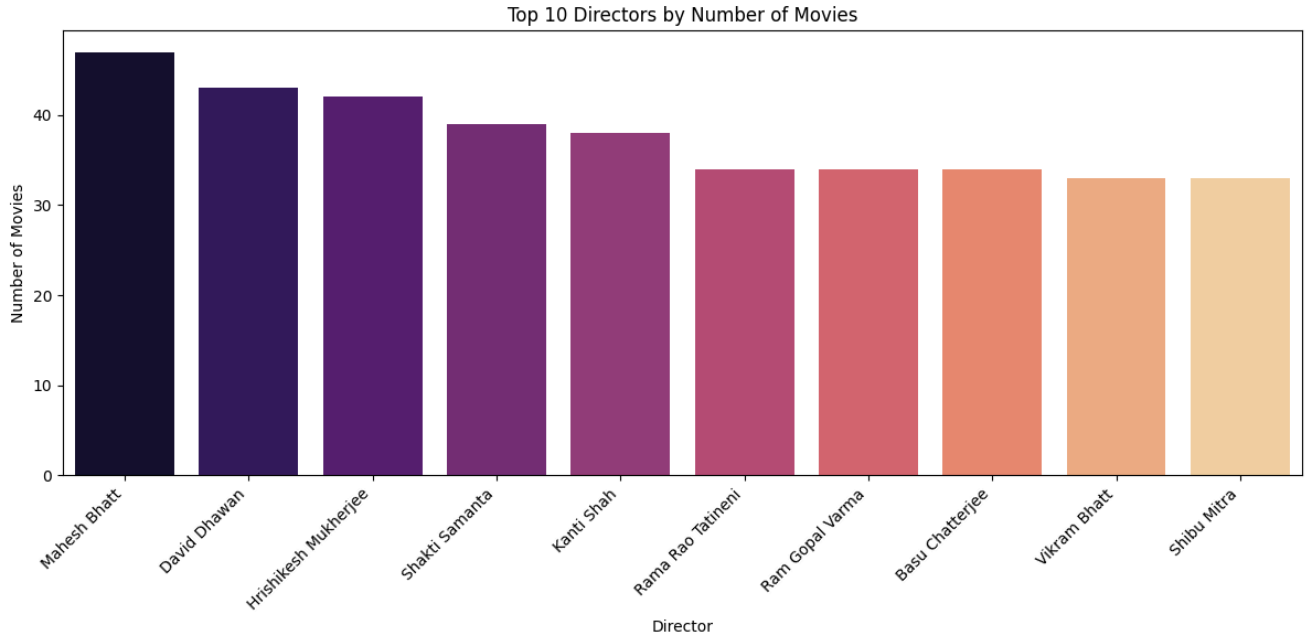

```
avg_rating_by_year = data.groupby('Year')['Rating'].mean()

plt.figure(figsize=(10, 5))
sea.lineplot(x=avg_rating_by_year.index, y=avg_rating_by_year.values, color='purple')
plt.title('Average Movie Rating by Year')
plt.xlabel('Year')
plt.ylabel('Average Rating')
plt.show()
```




```
# Get top 10 directors by number of movies
top_directors = data['Director'].value_counts().head(10)

# Plot
plt.figure(figsize=(12,6))
sea.barplot(x=top_directors.index, y=top_directors.values, palette='magma')

# Beautify
plt.xticks(rotation=45, ha='right')  # Rotate director names for readability
plt.title('Top 10 Directors by Number of Movies')
plt.xlabel('Director')
```

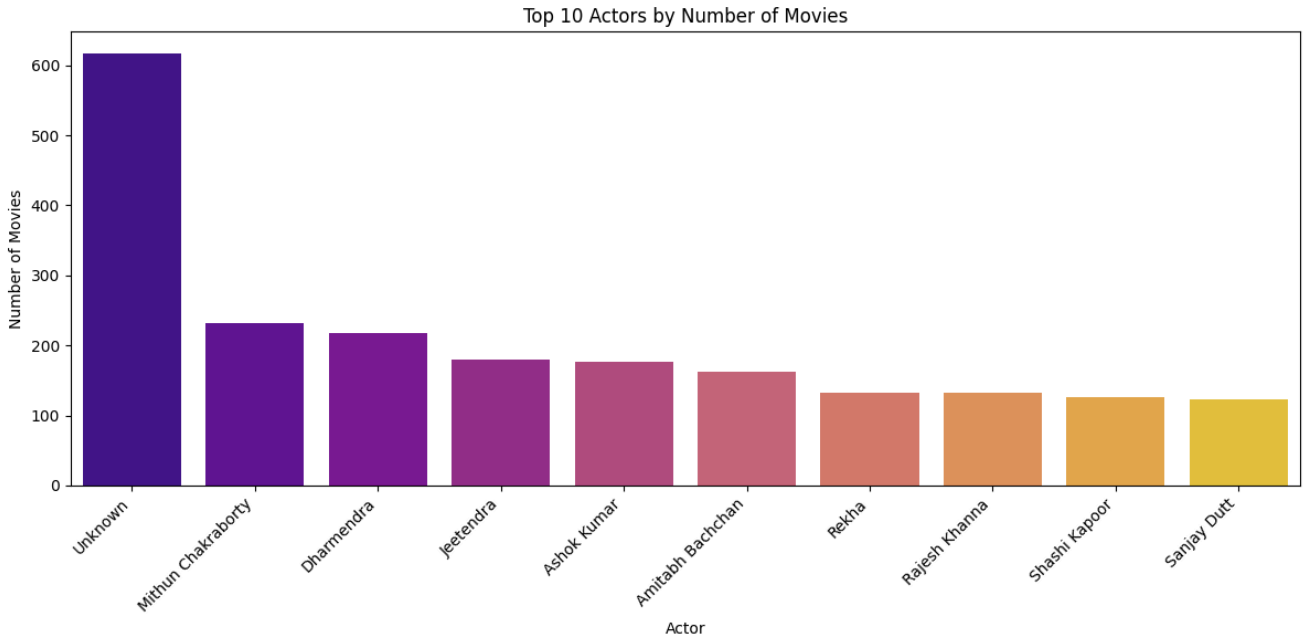
```
plt.ylabel('Number of Movies')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_1519/616335150.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sea.barplot(x=top_directors.index, y=top_directors.values, palette='magma')
```



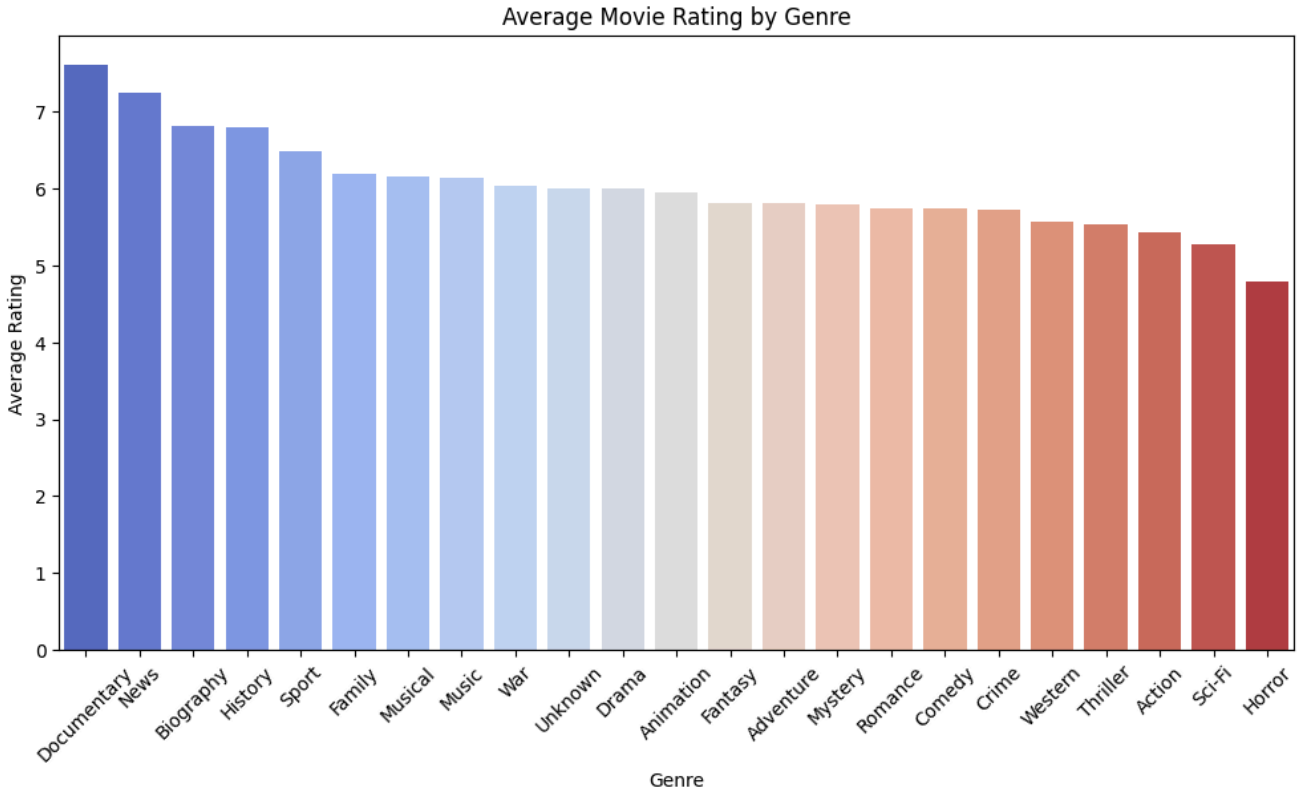

```
# Combine all actor columns into a single Series
all_actors = pd.concat([data['Actor 1'], data['Actor 2'], data['Actor 3']])

# Count appearances of each actor
top_actors = all_actors.value_counts().head(10)

# Plot
plt.figure(figsize=(12,6))
sea.barplot(x=top_actors.index, y=top_actors.values, palette='plasma')
plt.xticks(rotation=45, ha='right')
plt.title('Top 10 Actors by Number of Movies')
plt.xlabel('Actor')
plt.ylabel('Number of Movies')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_1519/160999834.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sea.barplot(x=top_actors.index, y=top_actors.values, palette='plasma')
```




```
avg_ratings = {}
for genre in genre_dummies:
    avg_ratings[genre] = data.loc[data[genre] == 1, 'Rating'].mean()

avg_ratings = pd.Series(avg_ratings).sort_values(ascending=False)

plt.figure(figsize=(12,6))
sea.barplot(x=avg_ratings.index, y=avg_ratings.values, palette='coolwarm')
plt.xticks(rotation=45)
plt.title('Average Movie Rating by Genre')
plt.ylabel('Average Rating')
plt.xlabel('Genre')
plt.show()
```

```
/tmp/ipykernel_1519/1826119636.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sea.barplot(x=avg_ratings.index, y=avg_ratings.values, palette='coolwarm')
```




```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7919 entries, 0 to 7918
Data columns (total 30 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Year         7919 non-null   float64
 1   Rating       7919 non-null   float64
 2   Votes        7919 non-null   float64
 3   Director     7919 non-null   object
 4   Actor 1      7919 non-null   object
 5   Actor 2      7919 non-null   object
 6   Actor 3      7919 non-null   object
 7   Action       7919 non-null   int64
 8   Adventure    7919 non-null   int64
 9   Animation    7919 non-null   int64
 10  Biography    7919 non-null   int64
 11  Comedy       7919 non-null   int64
 12  Crime        7919 non-null   int64
 13  Documentary  7919 non-null   int64
 14  Drama        7919 non-null   int64
 15  Family       7919 non-null   int64
 16  Fantasy      7919 non-null   int64
 17  History      7919 non-null   int64
 18  Horror       7919 non-null   int64
 19  Music        7919 non-null   int64
 20  Musical      7919 non-null   int64
 21  Mystery      7919 non-null   int64
 22  News         7919 non-null   int64
 23  Romance      7919 non-null   int64
 24  Sci-Fi       7919 non-null   int64
 25  Sport        7919 non-null   int64
 26  Thriller     7919 non-null   int64
 27  Unknown      7919 non-null   int64
 28  War          7919 non-null   int64
 29  Western      7919 non-null   int64
dtypes: float64(3), int64(23), object(4)
memory usage: 1.8+ MB
```

```
object_columns = ['Director', 'Actor 1', 'Actor 2', 'Actor 3']
# For each column, calculate the frequency of each name and map it back
for col in object_columns:
    # Count how many times each name appears
    freq_map = data[col].value_counts()

    # Map those counts to a new column
    data[col + '_freq'] = data[col].map(freq_map)

# Now drop the original string columns (optional, to keep things clean)
# data.drop(columns=object_columns, inplace=True)
```

```
print(data[['Director_freq', 'Actor 1_freq', 'Actor 2_freq', 'Actor 3_freq']].head())
```

```
   Director_freq  Actor 1_freq  Actor 2_freq  Actor 3_freq
0              1             2             1             1
1              1             5             1             2
2             17            18            15            13
3              7            25             4             8
4              1             1             6             1
```

```
data_model = data.drop(columns=['Director', 'Actor 1', 'Actor 2', 'Actor 3'])
```

```
# MODEL BUILDING
```

```
# importing modules
from sklearn.model_selection import train_test_split

X = data_model.drop('Rating', axis=1)  # All columns except 'Rating'
y = data_model['Rating']               # The column we want to predict

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()      # Create the model object
model.fit(X_train, y_train)     # Train (fit) the model on training data
```

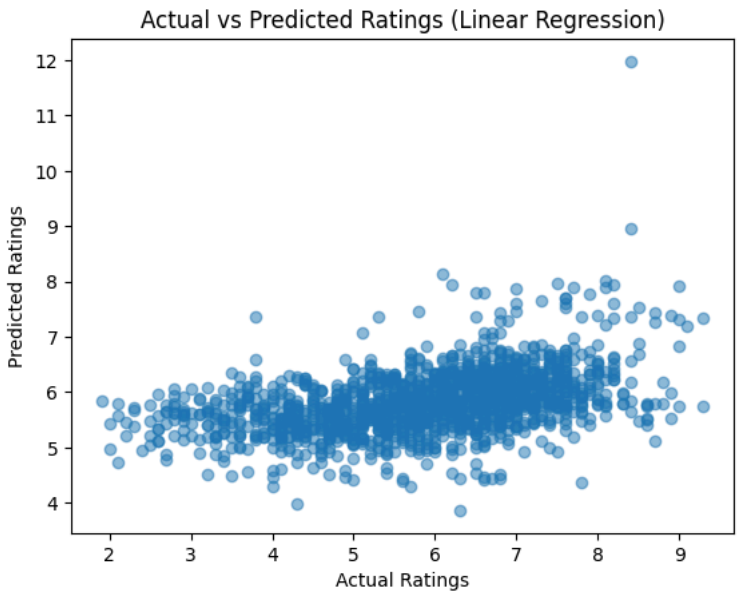
```
▾ LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X_test)  # Predict ratings for the test data
```

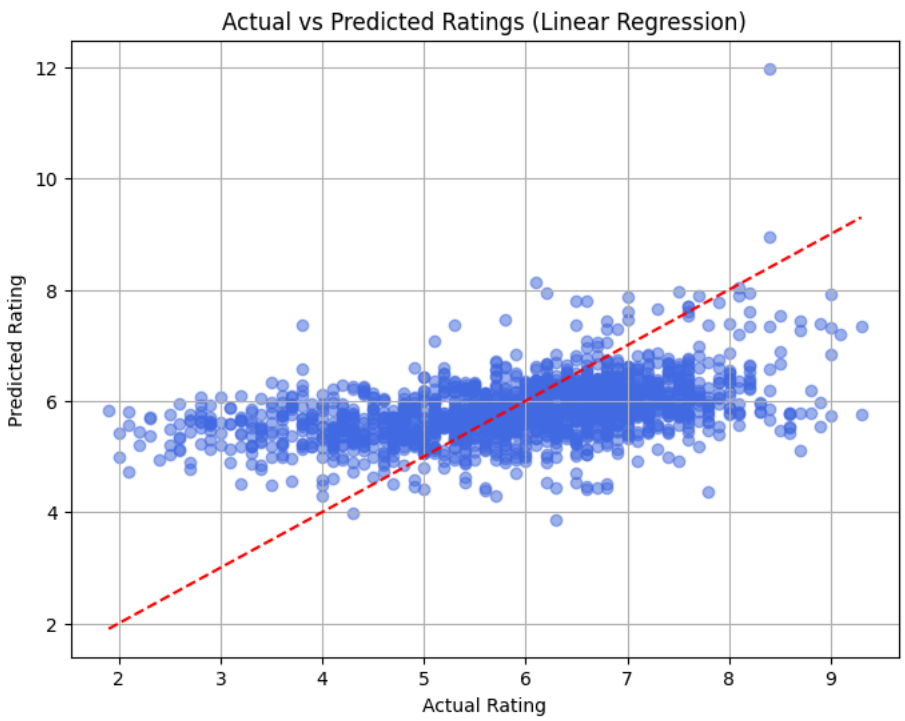
```
from sklearn.metrics import mean_squared_error, r2_score

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error:", mse)
print("R² Score:", r2)
```

```
Mean Squared Error: 1.5100886337922115
R² Score: 0.1877510793641899
```

```
plt.scatter(y_test, y_pred, alpha=0.5)
plt.xlabel("Actual Ratings")
plt.ylabel("Predicted Ratings")
plt.title("Actual vs Predicted Ratings (Linear Regression)")
plt.show()
```




```
# Scatter plot of Actual vs Predicted
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.5, color='royalblue')
plt.plot([min(y_test), max(y_test)], [min(y_test), max(y_test)], color='red', linestyle='--')  # Diagonal line
plt.xlabel("Actual Rating")
plt.ylabel("Predicted Rating")
plt.title("Actual vs Predicted Ratings (Linear Regression)")
plt.grid(True)
plt.show()
```




```
#Each dot = one movie.
#Red dashed line = perfect prediction.
#Dots closer to the red line = better predictions.
#Dots far from the line = model missed the mark
```

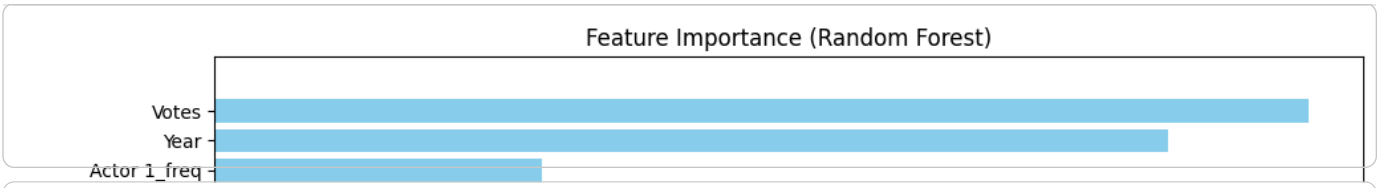
```
from sklearn.metrics import mean_absolute_error
import numpy as np

# Predictions already made as y_pred
```

```
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error (MAE):", round(mae, 3))
print("Mean Squared Error (MSE):", round(mse, 3))
print("Root Mean Squared Error (RMSE):", round(rmse, 3))
print("R² Score:", round(r2, 3))
```

```
Mean Absolute Error (MAE): 0.976
Mean Squared Error (MSE): 1.51
Root Mean Squared Error (RMSE): 1.229
R² Score: 0.188
```

```
from sklearn.ensemble import RandomForestRegressor

# Train the model
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

# Make predictions
rf_pred = rf_model.predict(X_test)
```

```
# Get feature importances
importances = rf_model.feature_importances_
feature_names = X.columns

# Create a DataFrame for easy sorting and plotting
feat_imp_df = pd.DataFrame({'Feature': feature_names, 'Importance': importances})
feat_imp_df = feat_imp_df.sort_values(by='Importance', ascending=True)  # ascending for barh plot

# Plot
plt.figure(figsize=(10, 8))
plt.barh(feat_imp_df['Feature'], feat_imp_df['Importance'], color='skyblue')
plt.title("Feature Importance (Random Forest)")
plt.xlabel("Importance Score")
plt.tight_layout()
plt.show()
```




```
# Calculate residuals
```